Claims

Claim 1

A dynamic pricing and verification system based on result validation, characterized in that the system comprises:

1. A **Price Limitation Gateway**: deployed in an edge reverse proxy server at the network access layer, configured to maintain a **three-tier gradient price gateway state machine** including a Trial Tier, a Paid LLM Service Tier and a Human Pay-by-Result Tier; it controls forced transition and interception of billing modes among the three-tier state machine through a cumulative invocation counter at the hardware socket level; when the Human Pay-by-Result Tier is triggered, it forces user nodes to send settlement pre-authorized escrow funds to the on-chain escrow account for locking;

2. A **Result Self-Discipline Validation Sandbox**: deployed in a local hardware trusted execution environment (TEE), configured to securely read unredacted private interaction slice feature vectors from a local database through a unidirectional physical tunnel when a session invocation occurs, and run **quantitative result assertion determination rules** encapsulated in a standardized protocol container to perform causal projection comparison on the feature vectors, so as to calculate quantitative result indicators; if and only if the result indicators reach the preset mathematical boundary threshold, it physically purges raw data including privacy details inside the isolated execution environment and generates a **desensitized Boolean assertion proof** that only indicates result achievement, and invokes the unique hardware-protected private key of the isolated execution environment to generate and output a cryptographic digital signature for the Boolean assertion proof;

3. A **Clearing Routing Contract**: deployed on a decentralized ledger, comprising a capacity monitoring module, a traffic shunt router and a settlement module; wherein the capacity monitoring module is configured to count the number of concurrent active human service sessions of each service Chuan Guang Shi in real time within the ledger execution context; when the number of concurrent active human service sessions of the target Chuan Guang Shi is greater than or equal to the preset service saturation capacity limit, it automatically triggers the traffic shunt router to pull a list of homologous sub-protocols with multi-generation fingerprint alignment, and shunts and routes the current session invocation traffic to the selected sub-protocol service node via a gateway redirection instruction; the settlement module receives the desensitized Boolean assertion proof broadcast by the Result Self-Discipline Validation Sandbox and verifies the signature using

the corresponding public key; upon successful signature verification and a determination of result achievement, it triggers programmable payment directly connected to multi-party digital wallets; upon a determination of unachieved result, it triggers a fee exemption state machine transition and transfers the locked escrow fund balance in the escrow account back to the user's address.

Claim 2

The verification system according to **Claim 1**, characterized in that:
The settlement module adopts a **pull-based withdrawal topology** during clearing and settlement; after calculating the allocated funds for each participant, the settlement module only updates the pending withdrawal balance state mapping variables corresponding to each participant on the ledger, without executing proactive fund pushes. Each participant node triggers fund withdrawal by sending an external withdrawal transaction, so as to physically block contract concurrent reentrancy and abnormal Gas overflow from the system architecture level.

Claim 3

The verification system according to **Claim 1**, characterized in that:
After the traffic shunt router redirects invocation traffic to the homologous sub-protocol service node, if the sub-protocol triggers pay-by-result payment, the settlement module automatically allocates a fixed proportion of the transaction revenue to the account address of the owner of the main value protocol as royalty compensation for copyright authorization.

Claim 4

A method for dynamic pricing and verification based on result validation, characterized in that, executed by a computer processor, the method comprises the following steps:
1. Price gateway transition control: a Price Limitation Gateway deployed in an edge reverse proxy server at the network access layer maintains a three-tier gradient price gateway state machine including a Trial Tier, a Paid LLM Service Tier and a Human Pay-by-Result Tier, and controls forced transition and interception of billing modes among the three-tier state machine through a cumulative invocation counter at the hardware socket level; when the Human Pay-by-Result Tier is

triggered, it forces user nodes to send settlement pre-authorized escrow funds to the on-chain escrow account for locking;

2. Human service saturation traffic routing: a Clearing Routing Contract monitors the number of concurrent active human service sessions within the ledger execution context; when the number of concurrent active human service sessions of the target Chuan Guang Shi is greater than or equal to the preset service saturation capacity limit, it automatically triggers the traffic shunt router to pull a list of homologous sub-protocols with multi-generation fingerprint alignment, and shunts and routes the current session invocation traffic to the selected sub-protocol service node via a gateway redirection instruction;

3. Sandbox quantitative result validation: a Result Self-Discipline Validation Sandbox deployed in a local hardware trusted execution environment securely reads unredacted private interaction slice feature vectors from a local database through a unidirectional physical tunnel when a session invocation occurs, and runs quantitative result assertion determination rules to perform causal projection comparison on the feature vectors to calculate quantitative result indicators; if and only if the result indicators reach the preset mathematical boundary threshold, it physically purges raw plaintext data inside the isolated execution environment, generates a desensitized Boolean assertion proof, and outputs it after signing with a hardware private key;

4. Clearing execution and exemption: the on-chain contract receives and verifies the digital signature proof to confirm it originates from an authorized sandbox; upon successful signature verification and a determination of result achievement, it triggers programmable payment directly connected to multi-party digital wallets; upon a determination of unachieved result, it triggers a fee exemption state machine transition and transfers the locked escrow fund balance in the escrow account back to the user's address.

Claim 5

The method according to **Claim 4**, characterized in that:

In the price gateway transition control step, the gradient transition is specifically as follows: when the cumulative free trial count **C_trial** exceeds the free quota limit and 1 ≤ C_trial ≤ 3, the state machine transitions from the Trial Tier to the Paid AI Service Tier; when the cumulative paid AI service count **C_paid** exceeds the limit of 3, the state machine forcibly transitions from the Paid AI Service Tier to the Human Pay-by-Result Tier, mandating the user to send settlement pre-authorized pledge funds to the clearing contract.

Claim 6

The method according to **Claim 4**, characterized in that:
In the clearing execution and exemption step, the execution mechanism of direct e-CNY (digital Renminbi) programmable payment includes: the on-chain settlement module first locks the current session's ledger pending withdrawal balance mapping state, invokes the programmable payment API interface connected to systems of secondary operating institutions of the central bank through a distributed secure socket, and triggers peer-to-peer clearing and transfer between the personal or enterprise digital fiat currency wallets bound to both transaction parties; if and only if a transaction success receipt signal returned by the secondary operating institution system is received, it releases the lock on the ledger state and clears the corresponding escrow pre-authorization counter within constant time complexity, so as to block asynchronous concurrent reentrancy between heterogeneous clearing gateways through a state lockstep mechanism.

Claim 7

A ledger contract-side operation method in the dynamic pricing and verification settlement method based on result validation, characterized in that, executed by a clearing routing contract deployed on a decentralized clearing and settlement ledger, the method comprises the following steps:
1. Session gateway transition control: monitor the establishment of invocation sessions and changes of the Price Limitation Gateway counter, implement three-tier state transition control of trial, paid AI and human services through hardware socket interception, and receive and lock the user's pre-authorized escrow funds;
2. Session activity saturation monitoring: monitor the number of active service sessions of the target Chuan Guang Shi; when it is greater than or equal to the preset service saturation capacity limit, trigger a traffic shunt mechanism to redirect and shunt the current session traffic to its registered homologous sub-protocol service nodes;
3. Desensitized assertion signature verification and settlement: receive the cryptographic result assertion signature proof broadcast by the external sandbox and verify the signature using a public key; when the signature verification passes and the assertion is true, lock the ledger mapping state, trigger the programmable API interface connected to the e-CNY system to execute transfer, and clear the escrow pre-authorization counter to release the lock after receiving a success receipt;
4. Result non-compliance exemption and refund: when the signature verification passes and the assertion is false, determine that the service fails to meet the standard, release the frozen escrow funds and transfer the balance back to the user's address.

Claim 8

A result validation sandbox-side operation method in the dynamic pricing and verification settlement method based on result validation, characterized in that, executed by a secure validation microcontroller configured inside a local result self-discipline validation sandbox, the method comprises the following steps:

1. Local result validation and sandbox loading: securely read unredacted private interaction slice feature vectors through a unidirectional physical tunnel when a session invocation occurs, and run quantitative result assertion determination rules;

2. Result indicator projection comparison: perform causal projection comparison on private interaction slice feature vectors inside the sandbox to compare them with pre-allocated quantitative result indicators;

3. Desensitized assertion generation and signing: if and only if quantitative result indicators reach the mathematical boundary threshold, generate a desensitized Boolean assertion proof indicating result achievement, and output a signature proof after signing it with a hardware private key.

Claim 9

The decentralized clearing contract-side operation method according to **Claim 7**, characterized in that:

The session gateway transition control step further includes a temporal random number and digital signature verification sub-step to prevent replay attacks; and in the desensitized assertion signature verification and settlement step, when locking the ledger mapping state, a concurrency control state identifier based on optimistic locking and concurrent version control is added to handle multi-party concurrent settlement requests.

Claim 10

The local result validation sandbox-side operation method according to **Claim 8**, characterized in that:

In the sandbox private key digital signature step, the latest block hash of the current session or a one-time random Nonce is mixed in when performing digital signature with the private key, so as to realize signature anti-replay; and before the encrypted proof file is sent to the decentralized

ledger, physical transmission encryption is performed through an asymmetric public key encryption channel to prevent man-in-the-middle interference and interception snooping.

Claim 11

A computer-readable storage medium having a computer program stored thereon, characterized in that the program, when executed by a processor, implements the method according to any one of **Claim 4** to **Claim 10**.